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Acute toxicity assessment nickel to freshwater air breathing teleost *Channa gachua*

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Abstract- Heavy metals are naturally occurring elements that have a density at least five times that of water and a high atomic weight. They are widely dispersed throughout the environment due to their many industrial, domestic, agricultural, medical, and technical uses, which begs the question of what effects they might have on the environment and human health. During the experiment, the freshwater fishes *Channa gachua* were exposed separately to different doses of nickel ($NiSO_4$) for 24 hrs, 48 hrs, 72 hrs and 96 hrs. At first, we determined median lethal concentration (LC_{50}) of $NiSO_4$ to fish using graphical interpolation method and confirmed it by probit analysis. Changes in locomotory and behavioral attributes were observed carefully and monitored properly. The treated fishes showed restlessness and rapid random movements. Rapid opercular movements were also observed.

Keywords: *Channa gachua*, lethal concentration, graphical interpolation method, probit analysis, Nickel

INTRODUCTION

The heavy metal contamination of water sources has become a significant global environmental issue, threatening aquatic ecosystems and human health due to their bioavailability and toxicity in various environmental components.[1] Over the past three decades, numerous researches have been carried out to assess the behaviour, destiny, and potential impacts of different heavy metals in the environment.

The quality of freshwater in both ground and surface systems is a significant concern, as consumable water must possess suitable mineral content. Extensive concentration of heavy metals has significant impacts on aquatic organisms. Usually, the main source of heavy metals are effluents from urbanization, industrialization, and agricultural activities in addition to electroplating and metal processing industries.[2,3]

Industrial wastewater effluents contain a wide range of hazardous HMs (heavy metals) pollutants, and the primary cause of increased environmental toxicity is human and anthropogenic impacts.[4,5] Natural activities including wind-driven soil erosion, wildfires, volcanic eruptions, biogenic processes, and sea salt discharges can naturally transfer heavy metals into ecosystems.[6] Mining activities, use of fertilizers, herbicides and pesticides, as well as irrigation of agricultural lands with untreated sewage and industrial effluents are some examples of anthropogenic contamination of the environment by heavy metals.[7-9]

Numerous research articles reveals that at least 80% of municipal wastewater is universally released into water bodies. Untreated and industrial waste is also responsible for dumping millions of tonnes of heavy metals, toxic

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sludge, many types of solvents and other effluents into water bodies annually.[10,11] Agriculture consumes "70% of the world's water which contributes significantly to water pollution".[11]

Nickel (Ni) is the fifth most prevalent elements on Earth. This metallic element has an appearance of silvery-white found in Earth's crust and core, as well as soil, water, and meteorites.[12] Being an integral part of several enzymes important for the cycling of carbon, nitrogen, and oxygen, it is considered important to both microorganisms and plants. From a scientific point of view, the presence of nickel (Ni) metal in animals is practically considered at essential basis; because, mammals who consume Ni-deficient diet leads to malnutrition.[13]

MATERIALS & METHODS

Healthy fishes, *Channa gachua*, of weight 40-45 grams and length 20-25 cm were collected from Patna and Haripur, in July, 2023 and were brought to the Patna Science College, Department of Zoology, Patna University, Patna. They were placed in plexi glass aquarium of 40L capacity and given a both of 0.04% $KMNO_4$ solution. Acclimatized in the laboratory for 15 days during which water was changed every 24 hours and they were given food ad libitum at 2- 3% of body weight per day. To the determination of LC_{50} of Nickel, stock solution of nickel sulphate ($NiSO_4$) was prepared by dissolving appropriate amount of $NiSO_4$ as Ni-salt in distilled water. Fishes were exposed to various concentration of nickel ($NiSO_4$) for 24, 48, 72, and 96 hrs. Simultaneously, the control group was also maintained.

Laboratory treated group of *C.gachua* fish were exposed to different/lethal concentration of nickel sulphate as: 150ppm, 300ppm, 600ppm, 1200ppm and their present survival at different times interval (24, 48, 72, and 96 hours) were recorded as a measure of acute toxicity test. The Statical analysis of all collected data were plotted statistically into regression lines (mortality in probity/ concentration i.e., probity exterminate/ concentration (LC_{50}) for 24, 48, 72, and 96 hrs. The mortality of the fish was recorded of the end of each exposure and firstly through Graphical Interpolation method LC_{50} was calculated, later it was confirmed by Probit Regression Analysis.

RESULTS

The present study highlights the proportional increase in the mortality of fishes with increasing concentration of Nickel. A linear relationship has been obtained between logarithm of concentration of Nickel and logarithm of fish mortality (Probit) for almost all the groups of treated fishes.

The calculated values of LC_{50} of $NiSO_4$ as Ni-salt for 24 hours, 48 hours, 72 hours and 96 hours were 150ppm, 300ppm, 600ppm, 1200ppm respectively by probit regression method. The regression equation mentioned in the Table I highlights that with the increase in period of exposure the mortality of fishes increased. Behavioural and physiological responses of the fishes also varied during the exposure. At the beginning of the experiment, fishes exhibited random movements and were seen to be restless. Later, they showed abnormal movements and behavioural response by moving haphazardly i.e., irregularly changing their directions. The rate of movement of operculum also increased appearing that they were trying to engulf more and more air. With passage of time these changes in responses became occasional.

Physico-chemical factors of water like temperature, alkalinity, hardness, pH, DO etc. have been reported to affect the extent of toxicity caused by $NiSO_4$.

Table 1- Data shows LC_{50} value of nickel as $NiSO_4$, for a period of 24, 48, 72, and 96 hrs. in freshwater fish *C. gachua*.

Aquarium No.	No. of test fish *C. gachua*	Nickel concentration	Mortality During							
			24hrs.		48hrs.		72hrs.		96hrs.	
			No.	%	No.	%	No.	%	No.	%
Gr. I	10	Control	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil
Gr. II	10	1200ppm	6	60	8	80	9	90	10	100
Gr. III	10	600ppm	4	40	6	60	6	60	7	70
Gr. IV	10	300ppm	2	20	3	30	4	40	5	50
Gr. V	10	150ppm	Nil	Nil	1	10	3	30	3	30
Gr. VI	10	110ppm	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil
Gr. VII	10	60ppm	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil

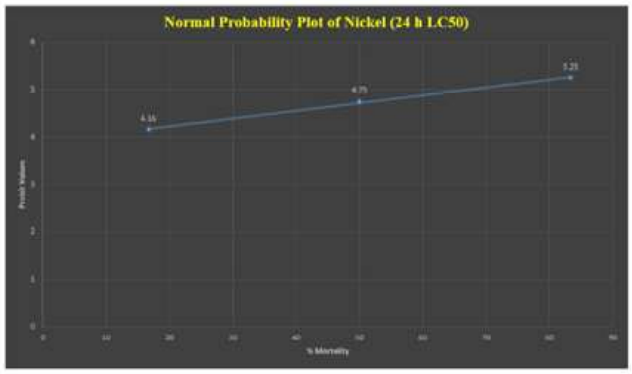


Fig:-Regression line based on probit analysis of $NiSO_4$ vs probit of percent mortality of *Channa gachua* at 48 hours

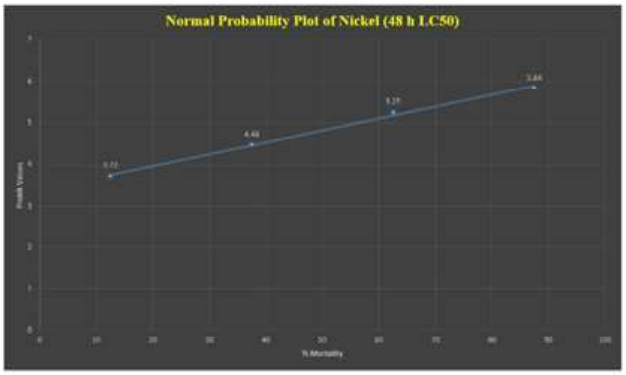


Fig:-Regression line based on probit analysis of $NiSO_4$ vs probit of percent mortality of *Channa gachua* at 48 hours

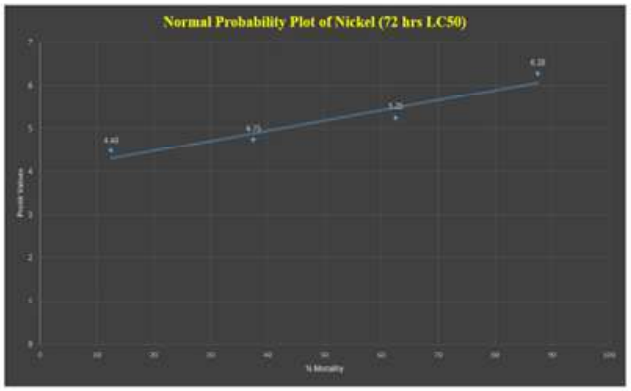


Fig:-Regression line based on probit analysis of $NiSO_4$ vs probit of percent mortality of *Channa gachua* at 72 hours

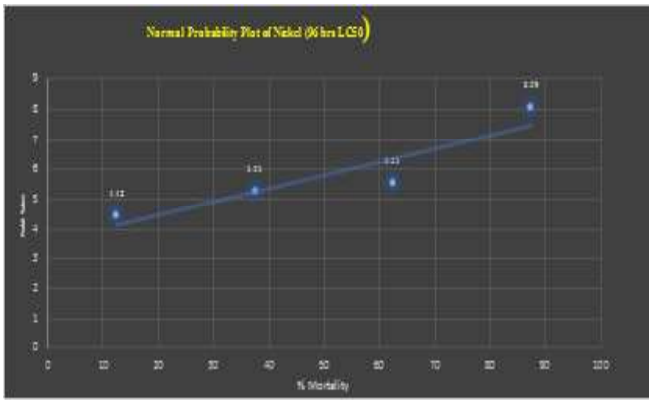


Fig:-Regression line based on probit analysis of $NiSO_4$ vs probit of percent mortality of *Channa gachua* at 96 hours

CONCLUSION

The fish *Channa gachua* were exposed to different concentrations of Nikel Sulfate and the LC_{50} values were calculated at 24 hours, 48 hours, 72 hours and 96 hours. We found here high values of LC_{50} which might have been due to various factors. Further studies are needed to understand the key factors that are playing vital role in variations in the LC_{50} values in *C. gachua* 50%

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